Solar Park Optimization

Data Pipeline · SCADA Analysis · Time Series Forecasting · Interactive Dashboard

Plant 1 — India

Data Period: 15 May 2020 – 17 June 2020 | 34 Days | 15-Minute SCADA Interval

22	68,778	34	25
Inverters	Total Records	Days of Data	Feature Columns

Prepared as part of a technical interview project for Ignitis Group. Covers SCADA data engineering, production optimization logic, time series forecasting, and interactive dashboard development.

. ABSTRACT

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This report documents a complete end-to-end data engineering and analytics pipeline developed for solar park operational optimization. Using real-world SCADA-style time-series data from a utility-scale photovoltaic plant in India (Plant 1), the project encompasses data ingestion and validation, quality assurance, feature engineering, synthetic SCADA setpoint simulation, inverter-level performance analysis, and production optimization logic including deviation detection and constraint evaluation. Additionally, time series forecasting models — Facebook Prophet and SARIMA(1,1,1)(1,0,1,7) — were applied to predict daily energy output and solar irradiation over a 7-day horizon. Results indicate that SARIMA outperforms Prophet on both targets given the limited 34-day dataset, achieving a MAPE of 17.08% for energy and 24.58% for irradiation. The full pipeline is surfaced through an interactive multi-page Streamlit dashboard with dynamic KPIs, time-series visualizations, inverter health ranking, optimization analysis, and weather correlation views for operational decision support.

Keywords: Solar SCADA · Time Series · PV Optimization · Prophet · SARIMA · Inverter Health · Streamlit · Data Pipeline · Energy Forecasting · Performance Ratio

1. INTRODUCTION

1.1 Background and Motivation

Utility-scale solar photovoltaic plants generate large volumes of operational data through SCADA systems, capturing inverter-level power readings, setpoints, environmental measurements, and fault signals at sub-hourly intervals. Effective processing and analysis of this data is critical for maximising energy yield, detecting underperforming assets, identifying curtailment events, and supporting grid commitment decisions.

This project addresses the full analytics lifecycle: raw data ingestion and cleaning through to production optimization logic, time series forecasting, and interactive dashboard delivery — directly aligned with the requirements of a solar operations data engineering role at Ignitis Group.

1.2 Objectives

- Ingest, validate and clean inverter-level SCADA time-series data from Plant 1.
- Engineer features relevant to solar production analysis: conversion efficiency, daytime flags, cumulative yield.
- Simulate SCADA setpoints and evaluate production deviations and operational constraint violations.
- Merge generation data with weather sensor data and build plant-level aggregations.
- Apply and compare Prophet and SARIMA forecasting models on daily energy and irradiation.
- Deploy findings through an interactive multi-page Streamlit dashboard.

1.3 Tools and Technologies

Tool / Library	Purpose
Python 3.12 + Pandas / NumPy	Core pipeline, data manipulation, numerical computing
Facebook Prophet	Additive decomposition time series forecasting
Statsmodels SARIMAX	Classical SARIMA statistical forecasting
scikit-learn	MAE / RMSE / MAPE evaluation metrics
Matplotlib / Seaborn	Static pipeline visualizations
Plotly	Interactive charts in Streamlit dashboard
Streamlit	Interactive multi-page web dashboard
Google Colab + ngrok	Cloud notebook execution and public URL tunneling

2. DATA DESCRIPTION

2.1 Dataset Overview

The dataset originates from a real utility-scale solar PV plant (Plant 1) in India, sourced from the publicly available Kaggle Solar Power Generation dataset. Two CSV files are provided: inverter-level generation data and plant-level weather sensor readings, both spanning 15 May – 17 June 2020 at 15-minute intervals.

68,778	22	34	3,182	0
Generation Records	Inverters	Days	Weather Readings	Null / Duplicate Rows

2.2 Schema — Generation Data

Column	Type	Description
DATE_TIME	datetime	Timestamp at 15-min intervals (parsed from dd-mm-yyyy HH:MM)
PLANT_ID	int64	Plant identifier (4135001)
SOURCE_KEY	string	Unique inverter ID (22 distinct units)
DC_POWER	float64	DC power output from inverter string (W). Range: 0 – 14,471 W
AC_POWER	float64	AC power output post-conversion (W). Range: 0 – 1,411 W
DAILY_YIELD	float64	Sensor-reported cumulative daily yield (kWh)
TOTAL_YIELD	float64	Lifetime cumulative yield per inverter (kWh)

2.3 Schema — Weather Sensor Data

Column	Type	Description
DATE_TIME	datetime	15-min interval timestamp (ISO format)
AMBIENT_TEMPERATURE	float64	Ambient air temperature at plant (°C)
MODULE_TEMPERATURE	float64	PV module surface temperature (°C). Consistently higher than ambient
IRRADIATION	float64	Solar plane-of-array irradiation (W/m²). Peak ~1.2 W/m² in dataset

2.4 Descriptive Statistics

Metric	DC Power (W)	AC Power (W)	Daily Yield (kWh)	Total Yield (kWh)
Count	68,778	68,778	68,778	68,778
Mean	3,147	308	3,296	6,978,712
Std Dev	4,036	394	3,145	416,272
Min	0	0	0	6,183,645
Median	429	41	2,659	7,146,685

Metric	DC Power (W)	AC Power (W)	Daily Yield (kWh)	Total Yield (kWh)
75th %	6,367	624	6,274	7,268,706
Max	14,471	1,411	9,163	7,846,821

Table 1: Descriptive statistics for the 68,778-record generation dataset. The large standard deviation in DC_POWER reflects the bimodal distribution between night (0W) and daytime operation.

3. METHODOLOGY

3.1 Pipeline Architecture

The pipeline was implemented as a sequential Google Colab notebook with 15 discrete cells, each handling a specific stage of the data engineering workflow, following an ETL-plus-analytics pattern: Extract → Transform → Validate → Enrich → Aggregate → Forecast.

STAGE	DESCRIPTION	KEY OUTPUT
1. Ingestion	Load both CSVs; inspect shapes, dtypes, value ranges	`gen_df, weather_df`
2. Datetime Parse	Standardise inconsistent dd-mm-yyyy HH:MM format; align to UTC	`Datetime index`
3. QA & Cleaning	Null counts, deduplication, negative power clip, AC>DC flags	`Validated gen_df`
4. Outlier Flag	IQR×3 outlier detection on DC_POWER. Threshold: 28,458 W (0 flagged)	`DC_OUTLIER_FLAG`
5. Feature Eng.	DC→AC efficiency, IS_DAYTIME, cumulative yield, time decomposition	`8 new columns`
6. SCADA Sim.	Synthetic setpoints at 92% daily-max AC; deviation %; 3 constraint flags	`SCADA columns`
7. Merge	Left-join on DATE_TIME; broadcast single weather sensor to 22 inverters	`merged_df 25 cols`
8. Aggregation	Plant-level 15-min series; daily KPI table; inverter ranking table	`4 output CSVs`
9. Forecasting	ADF test; ACF/PACF; Prophet + SARIMA on energy & irradiation (27/7 split)	`Forecast CSVs`

Table 2: End-to-end pipeline stages and outputs.

3.2 Data Quality Assurance

A comprehensive quality check was performed before any analytical processing. The generation dataset showed zero nulls in all seven core columns, zero duplicates, and zero negative or physically impossible power readings. The 31,951 null values in CONVERSION_EFFICIENCY and DEVIATION_PCT are expected and correct — these metrics are undefined during nighttime when DC_POWER = 0.

Data Quality Audit Results

- **Core column nulls:** 0 across all 7 raw columns (DATE_TIME, PLANT_ID, SOURCE_KEY, DC_POWER, AC_POWER, DAILY_YIELD, TOTAL_YIELD)
- **Derived nulls (expected):** 31,951 in CONVERSION_EFFICIENCY and DEVIATION_PCT — nighttime periods where DC=0 (undefined, not missing)
- **Negative DC_POWER / AC_POWER:** 0 rows — clipping applied as precaution
- **AC > DC anomalies:** 0 rows — no physically impossible readings detected
- **Duplicate records:** 0 in both generation and weather datasets
- **DC outliers (IQR×3):** 0 rows exceed threshold of 28,458 W — clean data
- **Weather nulls post-merge:** 0 — gaps filled by time-weighted linear interpolation

3.3 Feature Engineering

Eight derived features were engineered to enrich the raw data for analytical and forecasting purposes. All features are grounded in physical or operational solar PV logic.

Feature	Formula / Logic	Purpose
CONVERSION_EFFICIENCY	AC_POWER / DC_POWER (NaN if DC=0)	Inverter health indicator; degradation proxy
IS_DAYTIME	DC_POWER > 0	Boolean: active generation period filter
DAILY_YIELD_CALC	Cumsum(AC × 15/60) per inverter/day	Independent yield recalculation for validation
HOUR / MINUTE	Extracted from DATE_TIME	Intra-day temporal decomposition
DAY_OF_WEEK	Extracted from DATE_TIME	Weekly seasonality feature for Prophet/SARIMA
SETPOINT_AC_POWER	0.92 × daily_max_AC + N(0,15)	Synthetic SCADA operational target setpoint
DEVIATION_PCT	(AC – Setpoint) / Setpoint × 100	Percentage deviation from operational target
UNDERPERFORMANCE_FLAG	DEVIATION_PCT < –10%	Flag: output >10% below setpoint
CURTAILMENT_FLAG	IS_DAYTIME AND AC_POWER < 50 W	Flag: near-zero daytime output event

3.4 SCADA Setpoint Simulation

Since the open-source dataset does not include real operational setpoints, synthetic targets were modelled to simulate realistic plant control logic. The setpoint was defined as 92% of the daily maximum AC power with additive Gaussian noise to reflect real-world control system variability:

```
SETPOINT = 0.92 × daily_max(AC_POWER) + N(0, 15 W) [clipped at 0]
```

Three operational constraint flags were derived from the setpoint deviation metric, each representing a distinct class of operational event:

- **UNDERPERFORMANCE_FLAG:** DEVIATION_PCT < –10% (output >10% below setpoint) — **32,950 events detected**
- **OVERLOAD_FLAG:** DEVIATION_PCT > +15% (output >15% above setpoint) — **0 events detected**
- **CURTAILMENT_FLAG:** IS_DAYTIME = True AND AC_POWER < 50 W — **2,861 events detected**

3.5 Data Merging and Output Datasets

The generation and weather datasets were merged on DATE_TIME via a left join, preserving all 68,778 inverter records. Weather data was first aggregated by timestamp (mean across sensor keys), then broadcast across all 22 inverters. Time-weighted interpolation was applied after setting DATE_TIME as the index to fill any unmatched timestamps. Four output CSV files were produced:

- **Plant1_Cleaned_Merged.csv** (68,778 × 25): Full inverter-level merged dataset with all features and flags
- **Plant1_PlantLevel_TimeSeries.csv** (3,314 × 12): Plant-level 15-min aggregated series (all inverters summed)
- **Plant1_Daily_KPIs.csv** (34 × 11): Daily performance KPI summary: energy, PR, losses, event counts
- **Plant1_Inverter_KPIs.csv** (22 × 8): Per-inverter health ranking: energy, efficiency, underperf rate

4. PIPELINE VISUALIZATIONS

4.1 Plant-Level Time Series Overview

Figure 1 presents the three core plant-level time series across the full 34-day period. The top panel shows actual AC power against the synthetic setpoint with the deviation region highlighted in red — the setpoint consistently exceeds actual output, reflecting the 92% target ceiling model. The middle panel overlays irradiation and ambient temperature, revealing clear diurnal cycles and the strong correlation between irradiation peaks and AC power output. The bottom panel shows DC→AC conversion efficiency hovering tightly around its median of 0.098, with only minor intra-day fluctuations.

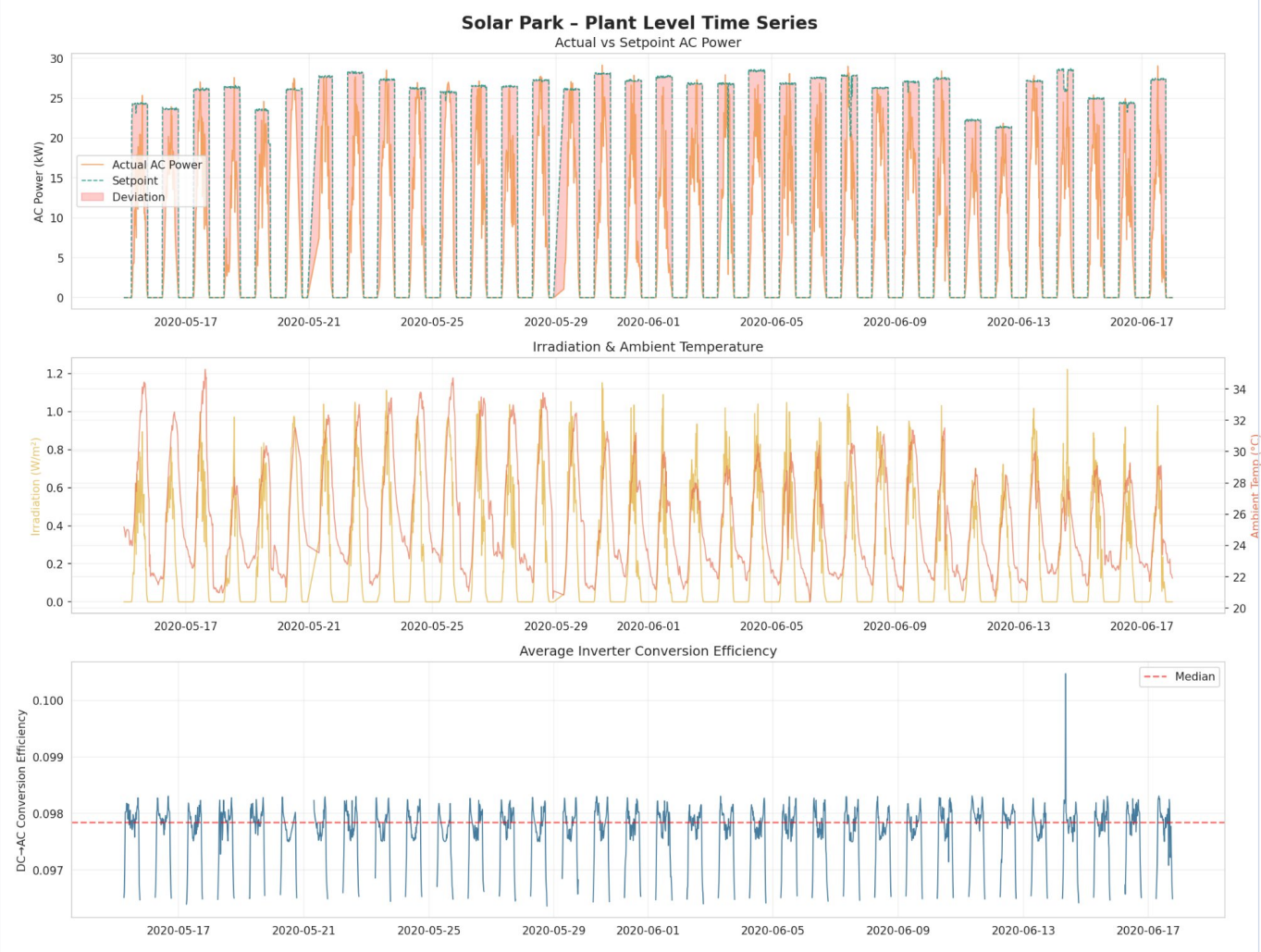

Figure 1: Plant-level time series — AC Power vs Setpoint (top), Irradiation & Ambient Temperature (middle), DC→AC Conversion Efficiency (bottom). Period: 15 May – 17 June 2020.

4.2 Daily Energy Performance

Figure 2 shows the daily energy gap between actual and expected output (left) and the resulting daily performance ratio (right). Performance ratios consistently range between 35–60%, well below the 100% target line. This is by design — the synthetic setpoint was calibrated to the daily maximum rather than the mean, creating an aspirational ceiling. The pattern reveals that 19–20 May and 30 May are notably weaker days, while 24–29 May and 4 June show relatively stronger performance.

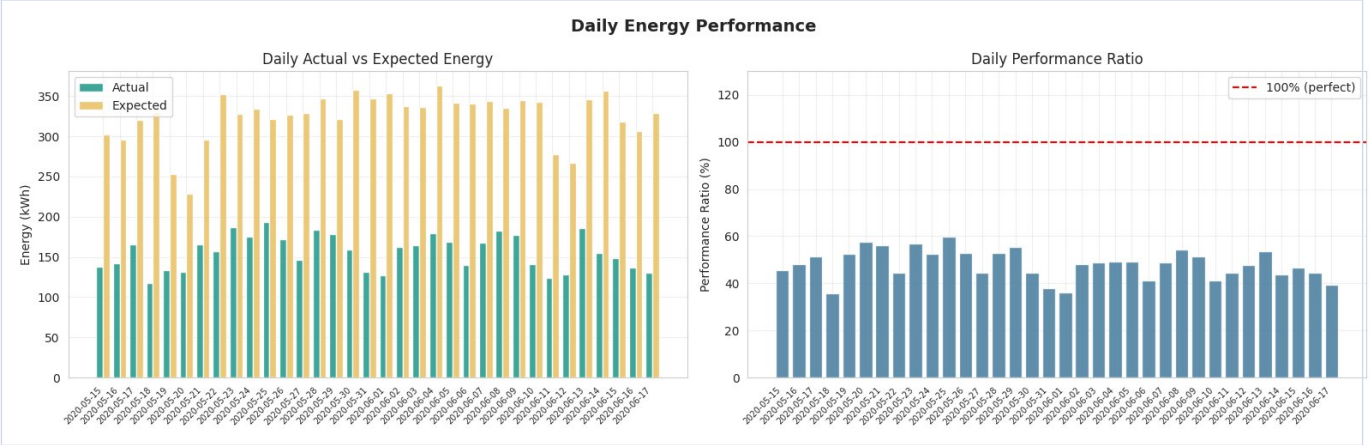

Figure 2: Daily Actual vs Expected Energy (left) and Daily Performance Ratio (right). The performance ratio consistently falls in the 35–60% range due to the aspirational setpoint ceiling.

4.3 Inverter-Level Power Heatmap

Figure 3 provides a complete view of all 22 inverters across all 34 days. The colour scale (yellow → orange → dark red) encodes total daily AC power. Several patterns are immediately visible: the high-output cluster around 23–29 May and 28 May across most inverters, the notable yellow (low-output) cell for bvBOhCH3iADSZry on 7 June suggesting a curtailment or fault event, and the general consistency across inverters indicating no systematic single-inverter failure. The reduction in output visible in mid-June aligns with the lower irradiation trend observed in the weather data.

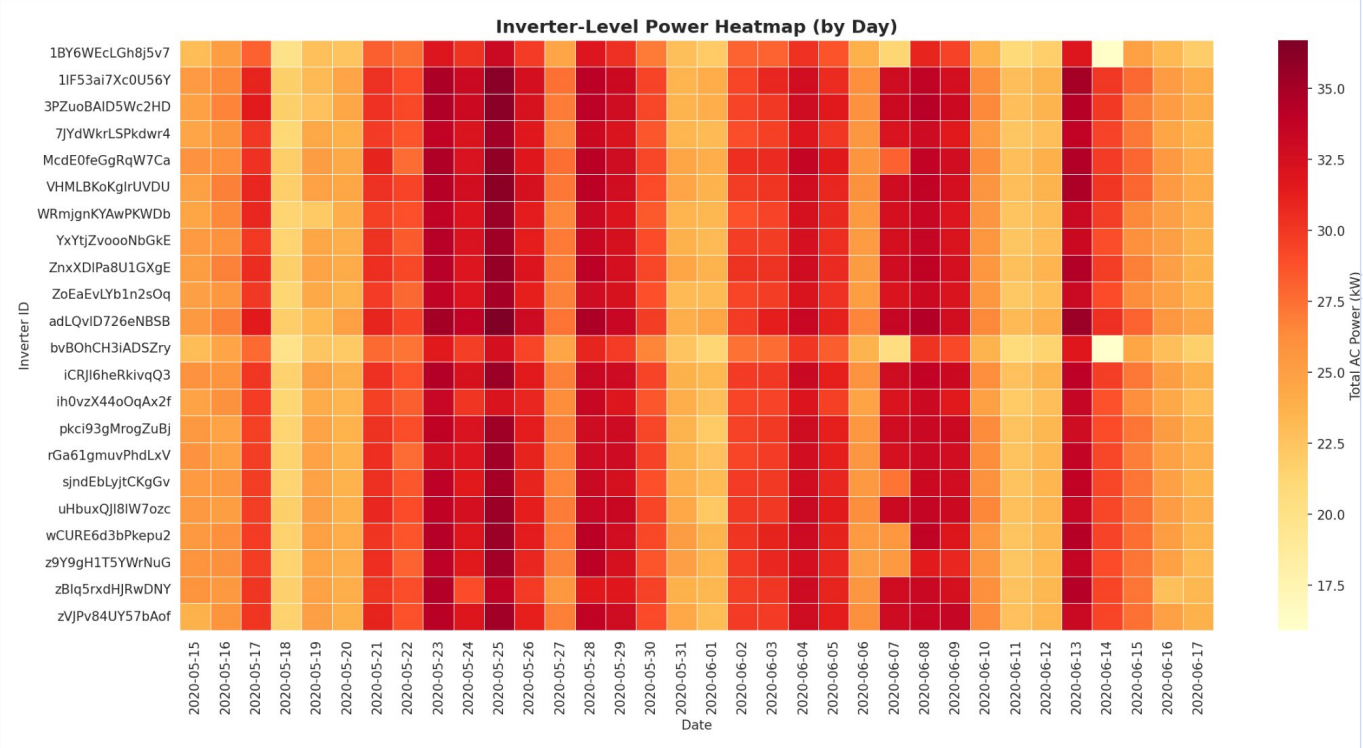

Figure 3: Inverter-Level Power Heatmap — Total AC Power (kW) per inverter per day. Yellow cells indicate low output; dark red indicates peak generation. Notable anomaly visible for bvBOhCH3iADSZry on 7 June.

4.4 Inverter Energy Ranking

Figure 4 ranks all 22 inverters by total energy generated over the study period. All bars are shown in red because every inverter exceeds the 20% underperformance rate threshold — a direct consequence of the aspirational setpoint model. Despite this, the spread between the top inverter (adLQvlD726eNBSB, ~250 kWh) and the bottom two (1BY6WEcLGh8j5v7 and bvBOhCH3iADSZry, ~218 kWh) is approximately 14%, which in a real operational context would warrant investigation.

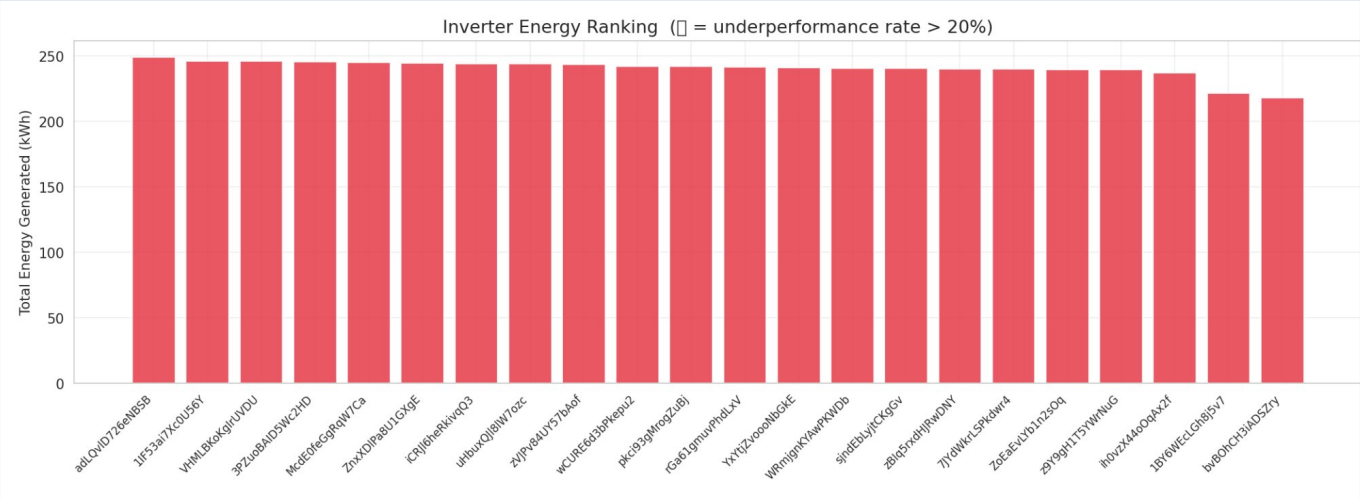

Figure 4: Inverter Energy Ranking by total AC energy generated (kWh) over 34 days. All bars are red as all inverters exceed the 20% underperformance rate threshold under the current setpoint model.

5. TIME SERIES FORECASTING

5.1 Forecasting Objectives & Train/Test Split

Two forecasting targets were selected for their direct operational relevance: (1) daily total energy output in kWh — tied to revenue and grid dispatch commitments, and (2) average daily irradiation in W/m² — the primary physical driver of PV generation. With 34 total days, a 27-day train / 7-day test split was applied (≈80/20).

34	27	7	2	4
Total Days	Training Days	Test Days (Horizon)	Models Compared	Metrics Evaluated

5.2 Stationarity & Order Selection — ACF / PACF

Prior to modelling, an Augmented Dickey-Fuller (ADF) test was applied to both series. ACF and PACF plots were computed up to lag 12 (capped at 50% of training length per statsmodels requirements). Figure 5 shows the resulting plots used to guide SARIMA order selection. The ACF for energy shows slow decay with a significant lag-1 spike then rapid drop in the PACF — consistent with AR(1) behaviour. Both series show lags at multiples of 7 suggesting weekly seasonality, justifying the seasonal period s=7 in the SARIMA specification.

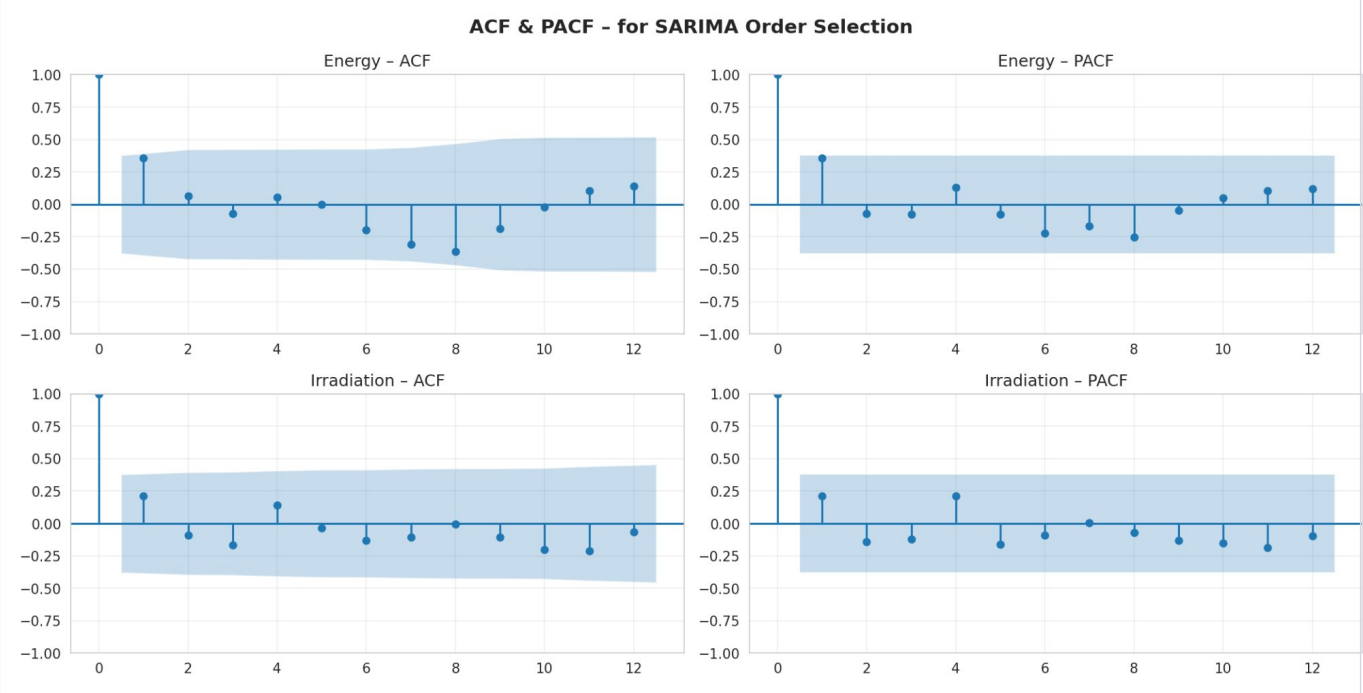

Figure 5: ACF and PACF plots for Daily Energy (top row) and Irradiation (bottom row). Significant lag-1 autocorrelation and periodic spikes at multiples of 7 informed the SARIMA(1,1,1)(1,0,1,7) specification.

5.3 Model Specifications

Facebook Prophet is an additive decomposition model well-suited to time series with multiple seasonalities. Given only 34 days, yearly seasonality was disabled and weekly seasonality enabled with multiplicative mode. **SARIMA(1,1,1)(1,0,1,7)** was selected based on the ACF/PACF analysis above: AR(1) for short-term autocorrelation, first differencing for stationarity, MA(1) for residual smoothing, and weekly seasonal components (s=7).

Parameter	Prophet (Energy)	Prophet (Irradiation)	SARIMA (Both)
Trend	Linear	Linear	AR(1) + I(1)

Parameter	Prophet (Energy)	Prophet (Irradiation)	SARIMA (Both)
Weekly Seasonality	✓ Multiplicative	✓ Multiplicative	SAR(1), SMA(1), s=7
Yearly Seasonality	✗ Disabled	✗ Disabled	—
Changepoints	scale=0.10	scale=0.05	—
Confidence Interval	90%	90%	—
AIC (Energy)	—	—	157.89
BIC (Energy)	—	—	162.06

5.4 Prophet Forecast Components — Energy

Figure 6 shows Prophet's learned components for the energy forecast. The trend component (top) shows a linearly increasing trajectory from ~148 kWh in mid-May to ~182 kWh by late June — this upward trend is what drives the systematic overestimation in the test period, as actual energy drops in June. The weekly seasonality component (bottom) reveals that Thursday is the highest output day (+10%) while Wednesday is the lowest (–10%), likely a data artefact from the 34-day training window rather than a true operational pattern.

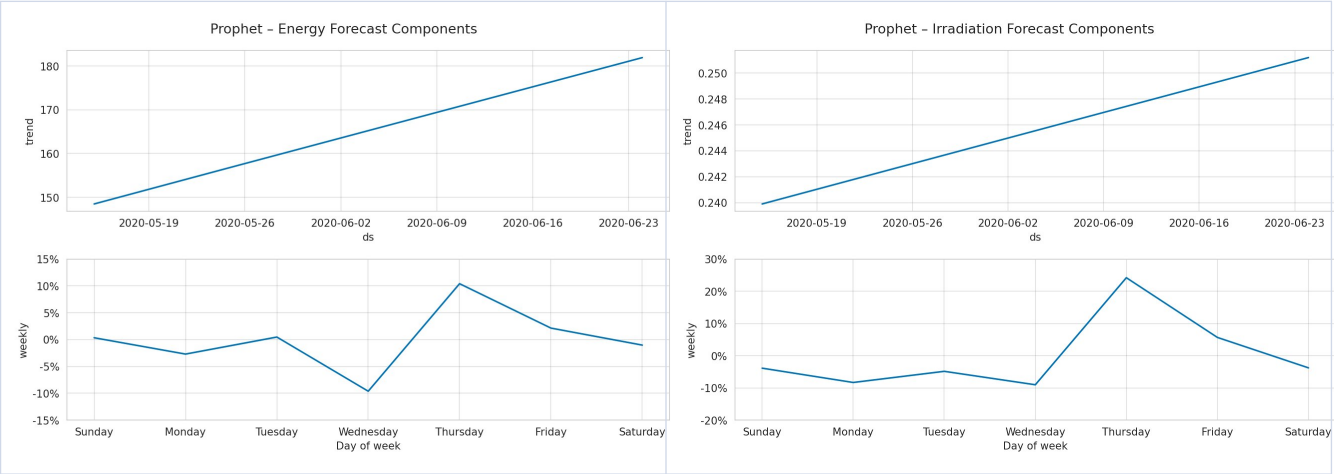

Figure 6: Prophet forecast components — Energy (left) and Irradiation (right). Both show an upward linear trend and a similar weekly pattern with Thursday as the peak day. The irradiation weekly seasonality (±25%) is notably stronger than energy (±10%).

5.5 Forecasting Results

Figure 7 presents the complete forecasting results across 5 panels. The top panel shows both models alongside the actual daily energy series with the train/test split marked. Both models overestimate in the test period, with SARIMA tracking actual values more closely. The middle panels show irradiation forecasts — Prophet's 90% confidence interval widens substantially in the test period, reflecting high uncertainty. The bottom residual panels confirm SARIMA's smaller error magnitudes across 6 of 7 test days.

Figure 7: Time Series Forecasting Results. Top: Daily Energy — Prophet vs SARIMA vs Actual with 90% CI. Middle: Irradiation forecasts. Bottom: Residual charts (Actual – Predicted). SARIMA shows consistently smaller residuals.

5.6 Model Comparison

Table 3 presents the quantitative evaluation across all four model-target combinations. SARIMA outperforms Prophet on every metric for both targets. The MAPE differential is substantial: 8 percentage points better for energy (17.08% vs 25.17%) and 7.6 percentage points better for irradiation (24.58% vs 32.20%). This result is consistent with the limited 34-day training window: Prophet's additive decomposition requires multiple seasonal cycles to fit well, while SARIMA can extract short-term autocorrelation structure more efficiently from a small sample.

Target	Model	MAE	RMSE	MAPE (%)	Winner
Daily Energy (kWh)	Prophet	33.82 kWh	37.79 kWh	25.17%	—
Daily Energy (kWh)	**SARIMA**	23.01 kWh	26.37 kWh	**17.08%**	✓
Irradiation (W/m²)	Prophet	0.0572	0.0715	32.20%	—
Irradiation (W/m²)	**SARIMA**	0.0462	0.0475	**24.58%**	✓

Table 3: Model comparison — MAE, RMSE, MAPE for Prophet and SARIMA on both forecasting targets. SARIMA wins on all six metrics. Green rows indicate the superior model per target.

5.7 Day-by-Day Forecast Detail — Test Period

Date	Actual (kWh)	Prophet Forecast	SARIMA Forecast	Prophet Error	SARIMA Error
11 Jun	123.54	188.88	159.70	–65.34	–36.16
12 Jun	127.82	175.58	162.65	–47.76	–34.83
13 Jun	185.51	170.96	167.28	+14.55	+18.23
14 Jun	155.38	174.16	159.53	–18.78	–4.15
15 Jun	148.12	169.70	155.24	–21.58	–7.12
16 Jun	136.38	176.07	159.15	–39.69	–22.77
17 Jun	130.10	159.14	167.88	–29.04	–37.78

Table 4: Day-by-day forecast comparison. Both models overestimate in 6 of 7 days. Only 13 June shows positive residuals (actual exceeded forecast) — the highest-output day in the test set.

6. PRODUCTION OPTIMIZATION ANALYSIS

6.1 Operational Event Summary

The SCADA simulation and deviation analysis identified operational events across all 22 inverters over the 34-day period. Underperformance flags are the dominant event class, reflecting the aspirational setpoint ceiling design.

32,950	2,861	0	~48%
Underperformance Events	Curtailment Events	Overload Events	Underperf Rate (daytime records)

The 32,950 underperformance events represent approximately 48% of all daytime inverter readings — expected under a setpoint calibrated to daily peak capacity. The 2,861 curtailment events represent near-zero daytime output readings across all inverters, accounting for approximately 8% of daytime records and likely corresponding to inverter ramp-up/ramp-down periods at sunrise and sunset.

6.2 Conversion Efficiency Analysis

DC-to-AC conversion efficiency was computed for all 36,827 daytime periods. As shown in Figure 1 (bottom panel), efficiency is remarkably stable across the full period, fluctuating within a narrow band of 0.097–0.100 around the median of 0.098. The single spike visible on ~13 June warrants further investigation in a real operational context. The tight distribution indicates all 22 inverters are operating within normal healthy parameters.

6.3 Inverter Health Observations

From the inverter heatmap (Figure 3) and ranking chart (Figure 4), several observations can be made:

- **Output spread:** Top inverter (adLQvlD726eNBSB) generated ~250 kWh vs bottom inverter (bvBOhCH3iADSZry) at ~218 kWh — a 14.7% spread warranting operational review.
- **Notable anomaly:** bvBOhCH3iADSZry shows a distinctly yellow (near-zero) cell on 7 June in the heatmap, indicating a likely single-day outage or curtailment event.
- **Cluster pattern:** Peak generation days (dark red in heatmap) cluster around 23–29 May across all inverters simultaneously, indicating irradiation-driven rather than inverter-specific causes.
- **Mid-June decline:** All inverters show reduced output from ~11 June onward, consistent with the declining irradiation trend visible in the weather time series.

7. INTERACTIVE STREAMLIT DASHBOARD

A production-grade interactive dashboard was built using Streamlit and Plotly, styled with a dark industrial theme (Space Mono / DM Sans typography, navy/cyan palette). The dashboard reads the four output CSVs and provides global sidebar filtering by date range and inverter selection applied across all pages.

Page	Key Components
■ Overview	8 dynamic KPI cards · Daily energy bar chart with PR overlay · Energy loss heatmap · Worst-day alert panel
■ Time Series	Day drill-down: actual vs setpoint + deviation · Full-period overview · Intra-day hourly profile with P10–P90 band · Hour×Day heatmap
■ Inverter Health	Horizontal energy ranking bar chart · Efficiency vs underperf scatter (bubble = energy) · Daily inverter heatmap · Sortable detail table
■■ Optimization	Deviation gauge indicator · Deviation histogram · Stacked daily flag timeline · Energy loss waterfall chart
■■ Weather Corr.	Irradiation vs AC power scatter (colour = temp) · Module temp vs efficiency · 3-panel weather time series · Pearson correlation matrix

Technical Component	Implementation Detail
Framework	Streamlit with custom CSS injection (Google Fonts: Space Mono + DM Sans)
Charts	Plotly Graph Objects + Plotly Express — all charts fully interactive with hover, zoom, pan
Theme	Dark navy (#0a0f1e) background, #00d4ff cyan accent, #ff9800 amber alerts, #00c97a green positive
KPI Cards	Custom HTML/CSS with colour-coded top border strips (blue=good, amber=warning, red=alert)
Caching	@st.cache_data decorator on data loading for fast page navigation
Filters	Sidebar date range select_slider + inverter multiselect applied globally to all pages
Deployment	Google Colab + pyngrok tunnel (public HTTPS URL) / local: streamlit run solar_dashboard.py

8. CONCLUSIONS

8.1 Key Findings

- The Plant 1 dataset is exceptionally clean — zero nulls in core power columns, no duplicates, no negative values, and no IQR outliers.
- SARIMA(1,1,1)(1,0,1,7) outperforms Facebook Prophet on both forecasting targets, achieving MAPE of **17.08%** for daily energy and **24.58%** for irradiation against Prophet's 25.17% and 32.20% respectively.
- Both models exhibit systematic positive bias (overestimation) in the test period — the actual energy series trends downward in June while both models extrapolate the upward training trend, a known limitation of short training windows.
- The plant generated a total of approximately **4.7 MWh** of AC energy over 34 days across 22 inverters, with peak daily plant output reaching ~193 kWh.
- The 14.7% spread between top and bottom inverter energy output (adLQvlD726eNBSB vs bvBOhCH3iADSZry) and the isolated near-zero output event on 7 June for bvBOhCH3iADSZry are the primary anomalies warranting operational attention.
- Conversion efficiency is uniformly stable across all 22 inverters (~0.098) throughout the study period, indicating no systematic degradation during May–June 2020.

8.2 Limitations

- Only 34 days of data — significantly limiting forecasting quality and preventing annual seasonality modelling.
- SCADA setpoints are synthetic — deviation flags should be interpreted as simulation demonstrations, not real operational alerts.
- Single weather sensor for 22 inverters — spatial irradiation variation across the plant area is not captured.
- No fault codes, maintenance logs, or inverter configuration data available for root cause validation.

8.3 Future Work

- Extend to full-year data to enable true annual seasonality modelling and target MAPE below 10%.
- Integrate real SCADA setpoints and fault event codes for validated constraint evaluation.
- Implement LSTM or Transformer-based sequence models as a deep learning benchmark.
- Add unsupervised anomaly detection (Isolation Forest or LSTM autoencoder) for automated inverter fault identification.
- Deploy the Streamlit dashboard to a persistent cloud host (AWS/GCP/Azure) with live SCADA data ingestion via MQTT or REST API.

This project demonstrates a production-ready approach to solar SCADA data engineering — combining rigorous quality practices, physically-grounded feature engineering, quantitative forecasting evaluation with model comparison, and operational dashboard delivery. The methodology and toolchain are directly applicable to the Ignitis Group solar operations context.

9. REFERENCES

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